

Locating Czech Constitutional Court Decisions in Doctrine Space and Measuring its Caselaw Consistency

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September 25, 2024

Abstract

Our research explores the estimation of positions of Czech Constitutional Court decisions in a doctrine space using Bayesian statistical model. Traditional methods of estimating ideological positions suffer from limitations, prompting the adoption of new text-as-data approaches empowered by advances in computational technology and statistics. Two research teams have attempted to overcome previous constraints and estimate judicial positions more accurately, one in the SCOTUS context and one in the German lower courts context. Our study implements the method of Clark and Lauderdale of estimating the locations of SCOTUS decisions with positive or negative references to its caselaw: the closer decisions are to each other, the more likely they are to cite themselves positively and vice-versa. We combine our own dataset of all CCC decisions with the data on citations provided to us by Beck. We use the programming language R and the Bayesian engine Stan to estimate the positions employing the Bayesian model of Clark and Lauderdale. Estimating the positions allows us to examine the consistency of the CC's case law across different senates and the plenum. We narrow our analysis to areas of law in common doctrine space that are prone to inconsistency, namely restitution cases and costs of civil proceedings. The research contributes to harnessing the potential of machine learning and quantitative methods in legal research and clarifies the factors influencing caselaw consistency.

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1 Introduction

The questions of how caselaw develops over time and to what extent inconsistencies in court case law come about have been of interest of legal scholarship as well as legal practice for quite some time. Different methods have been employed in to capture such development, ranging from simply legal interpretation (as an example [Vyhnanek 2023](#)), network analysis ([Lupu and Voeten 2012](#); [Coupette et al. 2021](#)), to adoption of item response theory models from psychometrics ([Clark and Lauderdale 2010](#); [Barberá 2015](#); [Arnold, Engst, and Gschwend 2023](#)).

In the presented paper, we deploy the scaling model based on the works of Clark and Lauderdale ([2010](#)); Barberá ([2015](#)); Arnold, Engst, and Gschwend ([2023](#)) in a three-fold fashion.

First, we employ it to uncover how the caselaw of the Czech Constitutional Court (CCC) develops over time. In the area of consumer cases, we discuss the importance of the so-called case-space in researching caselaw consistency. The example on consumer protection law also shows how the model, further developed in this article, can be employed to make topic modelling of court decisions more accurate.

Second, we employ the model to study the development of caselaw. We posit that the development in the form of precedents may be more gradual than previously assumed. Usually, a shift in court’s caselaw is equated with one landmark decision. Our data on the generally binding regulations (GBR) caselaw of the CCC shows that the shift may be more gradual and that it starts before the landmark decision.

Last, we take a jurisprudential article by Blažková ([2016](#)) concerning the right to effective investigation. We show that the claims made by Blažková can be and are supported by the data. Thus, we show that when a researcher comes up with a developed theory, our model can be employed to further back it with evidence that does not rely only on close reading of the caselaw.

In building upon the work, we fulfill the wish of Clark and Lauderdale ([2010](#)): “Given the centrality of theories of judicial policymaking to various substantive problems in political science, the method of scaling opinions developed in this article can facilitate a range of future research.” The development of computational technology enables employment of more advanced methods than simply reading the texts of court decisions, it does not render other methods useless. Quite on the contrary, our paper addresses many potential sources of struggle and limitations of employment of advanced quantitative methods, but it also shows how they may be employed in a constructive way.

Our article proceeds as follows: in [section 2](#), we discuss the relevant theoretical concepts of case-law consistency as well as of case-space. In [section 3](#), we discuss the method we employ. First, we describe the original scaling models as well as our adoption of thereof. We then discuss how

we practically narrowed the case-space down. Last, we describe the data we use. [Section 3](#) then discusses the results of our study. [Section 4](#) concludes.

2 Theory

The issue of estimating the development of caselaw overtime and its consistency effectively boils down to (1) establishing a consistent case-space, (2) estimating the location of a case in the case-space, and (3) assessing (quantitatively) its consistency. In all three aspects, we build on previous work - on theoretical work a common case-space ([Kornhauser 1992a, 1992b; Lax 2011](#)), on the empirical development of estimation of locations of court decisions ([Clark and Lauderdale 2010; Arnold, Engst, and Gschwend 2023](#)), and on theoretical work into consistency law across court chambers ([Fjelstul 2023](#)).

We now flesh out the theory for these subtopics. In [section 2.1](#), we discuss the concept and implications of case-space models. In [section 2.2](#), we discuss the concept of development and consistency of court caselaw. Lastly, in [section 2.3](#), we briefly go over the current work on estimating the location of court decisions. We reserve a more in depth discussion to [section 3](#) on method.

2.1 Case-space model

The case-space model was developed in an attempt to model the idiosyncrasies of court decision-making, i.e. that a court is a body resolving disputes, cases. Its starting point is that the idea of a case must be taken seriously: we must be able to represent a case and what rendering a judgment in any case means. In contrast to legislative models, “judicial preferences and goals are structured by the choices available to judges: dispositions and the rules that denote general sets of case dispositions. The second is that judicial choices, rule creation, and rule application are deeply affected by collegial and hierarchical (horizontal and vertical) divisions of power.” ([Lax 2011](#)) Thus, the case-space model incorporates into itself legal rules, the judges’ preference towards them, as well as the facts of the case.

More specifically, the way to represent a case is by locating it in a space of possible cases, the case-space. A case then denotes a “legally relevant event that has occurred” out of many that could have occurred. A car in a 30km/h zone could have in one case gone 28 km/h or in another 32 km/h. Both possibilities represent a case of a set of cases on the common case-space. A legal rule is, then, a partition of the case-space into whoever wins and whoever loses the case. However, where exactly the legal rule cuts across the case-space is determined by the judges’ preferences towards the legal rule. As [Lax \(2011\)](#) puts it, battles over law are struggles over partitions of the case space. The disposition of the case then refers to which party wins the case and results from the relationship between the judge’s preferences towards the legal rule and the facts of the case.

To be able to consistently estimate the location of CCC cases, they must all fall within the same case-space. In other words, they must be based on similar legal facts pertaining to similar legal rules that cut across the case-space. The development of caselaw overtime may for example shift the legal rule in any given case-space. It may be that with a precedent, the CCC sets a new standard for individual human rights review, which in all future cases makes the cases be more likely to be disposed of in favor of the applicants, as has been the case with consumer cases for example. How does then inconsistency of caselaw occur?

It is obvious that the real judicial adjudication is not as simple as a clear cut speeding rule. Typically, in order to resolve a case, the judges need to map the available evidence to the legal facts and then they have to select and interpret an appropriate legal rule to apply. Both processes are naturally subject to human judgment (Cameron and Kornhauser 2017). The indeterministic nature of this mapping opens up ways for disagreement. Landa and Lax (2007–2008) draw multiple theoretical sources of disagreement from the case-space model. The first and clear source of disagreement among judges is that about facts. Different judges may place the facts of the case in the case-space differently. There are other sources of disagreement: which dimensions should be relevant under any given legal rule to determine the disposition of a case, disagreement about “thresholds” within dimensions, and a couple of more sources of disagreement, which all can be summarized as a disagreement about the legal rule. All these sources of disagreement may lead to inconsistency across judges and chambers. If one judge views the facts of the case differently from others, a different chamber composed of different judges may not reach the same conclusion given the same set of legally relevant facts. The same applies to judges’ preferences over legal rules.

While the issue of development of caselaw and of development its inconsistency over time has been raised in the context of SCOTUS, or in the comparison between differing courts, such as Germany’s lower courts, we believe it to be equally applicable to collegial courts organised into multiple chambers, such as ECHR, CJEU or national apex (supreme and constitutional) courts. Therein, inconsistency can come about between the caselaw of different chambers of a collegial court. While works about caselaw consistency are typical doctrinal, presenting examples of mutually incompatible decisions, we study caselaw consistency empirically employing quantitative methods.

2.2 Case-law consistency

We now move on to elucidating how we understand the term *consistency* of case-law or of *a court’s application of the law* (Fjelstul 2023, 2) Our understanding builds upon the definition provided by Fjelstul in his theoretical study of the chamber system at the Court of Justice of the European Union.

Therein, Fjelstul theoretically conceptualizes *consistency* “as the degree to which the disposition of a case would change if it were decided by a different chamber” of a court. The theoretical definition is then translated into mathematical terms as “the variance of the distribution of the expected probability that the plaintiff wins a case across counterfactual chambers” (Fjelstul 2023, 2) Put differently, an application of law before a court is consistent only if a court would reach the same disposition with same facts, regardless of its composition or the deciding formation.

In theory, both the theoretical definition and operationalization are counter-factual as we can never observe how any given decision would be decided by more than one actual court chamber. We can, however, attempt to measure the trend across general population of cases falling into one case-space. And while we are aware that comparison based on observational data is not conclusive of any causal relationship, our ambition is only to offer a descriptive analysis of the CCC’s caselaw without necessarily uncovering any causal relationship in the sense of *what type of underlying feature leads to higher/lower consistency of CCC’s caselaw?*

Fjelstul’s operationalization of consistency assumes variation of chambers in a given point of time. Therefore, it also implicitly assumes that the legal situation is unchanged. If two chambers were to decide the same case at the same time differently, we might legitimately “see the judicial process

as arbitrary or capricious” (Fjelstul 2023, 2).

However, a court’s case law may also evolve in time. Although we may regard this process as undesirable too, it is almost inevitable in a large span of time with changing political and societal attitudes – or with changing the court’s composition. Therefore, looking at a totally disciplined court that respects its case law until the appropriate chamber (plenary, grand chamber) overturns it explicitly, we could describe its decision making as consistent in a given point of time (per Fjelstul’s definition), but still inconsistent in a span of time.

Thus, modeling consistency with a case dataset must take into consideration that the cases were decided at various points of time. This fact may distort the results, but at the same time, it enables us to study whether the case law has moved in any direction in a process of time. It can also help us to study the caselaw evolution through ignoring, distinguishing, limiting and explicitly rejecting earlier rulings. Spriggs and Hansford (2001), p. 1104 conclude that previous negative treatment of a precedent significantly increases the risk of its future overruling. Similarly, Reddick and Benesh (2000) show that lower courts may detect the case space direction SCOTUS is taking and even anticipate overruling its precedent. If that is the case, we should see in a given case space that decisions are incrementally moving in a certain direction, as opposed to one “big” decision that would make the switch. This may be even more visible in the institutional setting where only a designated court body (such as plenary session of the supreme court) can explicitly overrule precedent; then, such overruling will be probably preceded by past negative treatment (or ignorance) of the court’s individual chambers and/or lower courts.

2.2.1 The CCC and its case-law inconsistency

To estimate the location of cases in the case-space, we apply the model first devised by Barberá (2015) and then adopted into the legal context by Arnold, Engst, and Gschwend (2023). Based on these estimates, we will examine primarily the consistency of the Court’s case law across senates and the plenary and how it has developed over time. The interplay between the consistency of case law and a court’s internal structure deserves further inquiry.

The bulk of the CCC’s work is represented by individual constitutional complaints, which are, as a rule, decided upon by the senates; the judge rapporteur may declare certain complaints as inadmissible. The CCC decides in plenary sessions (*en banc*) mainly on review of primary and secondary legislation. The CCC rules either in a form of a judgment (*nález*) – i.e. a decision on merits – or in a form of a decision in a narrow sense (*usnesení*) where the complaint is rejected as inadmissible. A vast majority of the CCC cases are decided in the form of decision. As a legal rule, the CCC must respect its own case law, but only in the form of judgment (Kosař and Vyhnánek 2020, 161–63). A senate is obliged to submit a case to the plenary if it wants to overrule existing caselaw. Therefore, there is a formal mechanism enabling to heal inconsistencies of the CCC caselaw.

It follows from the internal structure of the CCC that each 3-member chamber might develop its “own” caselaw. Such caselaw, if in the form of judgment, would be binding for all public bodies, including the CCC itself. In theory, this (mal)practice is restricted by the necessity of submitting potentially inconsistent cases to the plenary. However, that requires a certain level of diligence and discipline on the side of the judges. It also does not prevent the senate from developing whole “new” caselaw, distinguishing it from earlier precedents if necessary rather than admitting

an inconsistency.

Justices may write dissenting and concurring opinions, but it is also possible to vote against the majority opinion without dissenting. Dissenting opinions are, therefore, an incomplete indication of disagreement within the CCC. That is seen by most authors as a major methodological problem in the empirical research of the CCC judges Vartazaryan (2022).

Scholarly literature is basically unanimous (Bobek and Kühn 2013) in the notion that the case law of the Constitutional Court is inconsistent due to the massive number of constitutional complaints (cf. Kosař and Vyhnánek 2020, 153), diffusion of decision-making bodies (cf. Kosař and Vyhnánek 2020, 154–55; Vyhnánek 2023, 15) and the conscious or unconscious failure to unify the case law. The 3-member senates either are unaware of conflicting case law, ignore it, or employ tricks to avoid its binding nature.

In the 3-member senate, the identity of judge rapporteur has such a significant effect on the probability that a constitutional complaint will be successful (Chmel 2021, 106–28) that the assignment of judge rapporteur has been described as “lottery” by other scholars (Chmel 2021, 129). This itself is a clear worrying sign from the complainant’s point of view. However, it does not necessarily imply caselaw inconsistencies. For instance, we could hypothesize that some justices are simply more “picky” (and vice versa) when deciding whether to issue a judgment but if they do, they decide in conformity with the caselaw. Still, given that the assignment of cases to judge rapporteurs is practically random, the varying winning chances reveal that the CCC judges have, on average, differing preferences regarding the disposition of a case.

Czech legal scholarship also provides examples of areas where the Constitutional Court’s decision-making practice is particularly inconsistent. It can be assumed that the level of inconsistency may be related to the value-based or legal-political dimension of the issue, which causes different judges to tend to decide differently. We go over a couple of them now.

The area of property restitutions (i.e. returning the property confiscated by the Communist regime to the original owners or compensating them) is widely considered to be one of the most disputed legal-political issues after the 1989 Velvet Revolution related to transitional justice, appearing many times before the CCC (Biagi 2020, 165–67). It is also frequently mentioned as the prime example of CCC’s inconsistency (Bobek and Kühn 2013, 363; Vyhnánek 2023). Surprisingly, restitution litigation was not restricted to the 1990s, often spanning decades – the neverending presence of the issue also helped by the restitution of church property enacted only in 2012 after a heated political and legal debate.

Our second area of interest is the consumer protection law. The area of consumer protection has enjoyed development only recently in the CCC caselaw, mainly thanks to the incorporation of EU law obligations. After all, the first decision from our dataset originated only in 2017. The caselaw originated in 2019. The second chamber of the CCC introduced EU law as one of the thresholds of constitutional review even in petty disputes (disputes in which the sum of its value does not exceed certain threshold) in consumer protection cases. Since then, similar cases have come up but they do not explicitly build on the previous caselaw.

The area of vertical separation of powers has undergone a longstanding development in the CCC caselaw. The Constitution enables territorial self-governing units (regions and municipalities) to issue legislation in the form of so-called “generally binding regulation” (*obecně závazná vyhláška*).

The CCC’s plenary has dealt with many cases where such regulation was under review. In the 1990s, the CCC applied a strict approach, ruling that a GBR needs an explicit statutory delegation and cannot in any way conflict with any Act of Parliament. These limits were eventually relaxed. In its landmark *Jirkov* judgment¹, the CCC stipulated that the power of issuing GBRs flows directly from the Constitution and municipalities therefore do not have to rely on any authorizing statute (GBR being, in that sense, primary legislation); and that a GBR may regulate an area which is already covered by a statute if the goals and purposes of both laws are different. This judgment was described by some as the “Copernican Revolution” of the CCC’s caselaw, while others downplayed its significance and noted that the CCC has already made steps in a more lenient direction in other cases (cf. [Holländer 2008](#); [Brož 2018](#)). In any case, the *Jirkov* judgment remains a respected precedent up-to-date. The question of a GBR’s legality, however, still often causes disagreement on the CCC benches.²

3 Method

Now that we explained the basic theoretical concepts as well as the areas of law we will study, we move on to discussing how we proceed practically. We start off with introducing how different researchers tackled the task of estimating location of court decisions. We then proceed to explain our model. We round off the section with introduction of the data employed.

3.1 Locating court decisions

Predominantly American empirical legal scholarship has long been occupied with estimating the position of judges or judicial decisions, especially in the context of political ideology of SCOTUS (Supreme Court of the United States). Based on this estimation, the research sought to answer subsequent sociological research questions. Spaeth-Segal ([Segal et al. 1995](#)) and Martin-Quinn ([Martin and Quinn 2002](#)) scores are among the examples of attempts to quantify the ideological position of judgments or judges.

However, these traditional estimation methods suffer from various shortcomings, such as measuring ideological positions indirectly (based on the party the decision was made against) or using data not available outside the American context (information about judges’ voting) and, thus, making the research near impossible to replicate elsewhere.

Recent developments in information technology enable more precise and diverse work with data, specifically with text of court decisions as data. Two research teams have utilised the progress to overcome the aforementioned limitations in traditional estimation efforts of judicial decisions and judges’ positions: Clark and Lauderdale ([2010](#)) and; Arnold, Engst, and Gschwend ([2023](#)). They employ Bayesian statistics to estimate the location of court decisions based on what legal sources they employ.

Both models rest on the assumption that a location of a court decision can be estimated based on its relationship to other court decisions and sources of law as courts justify their conclusion by reference to other court decisions or laws that are typically closer to them in the given case-space.

¹Judgment of the CCC of 11 December 2007, Pl. ÚS 45/06.

²For instance, five justices filed a dissenting opinion to the judgment of the CCC of 20 June 2023, Pl. ÚS 38/22, which struck down a GBR excluding too many nights within a calendar year from the statutory quiet hours.

The assumptions of such an approach are as follows: (a) each decision has a fixed location along a single latent dimensions delimited by a common case-space and (b) there is a proximity relationship, i.e. the probability of a (positive) reference to a court decision or other sources of law increases the closer the decisions are to each other in a case-space (Clark and Lauderdale 2010; Arnold, Engst, and Gschwend 2023).

The exact understanding of the relationship between a reference between two decisions and their position on the latent dimension can be understood in two manners. According to Clark and Lauderdale, *directional models* assume “that an opinion is more likely to cite a precedent if the precedent has the same [ideological] polarity as the opinion”, whereas *proximity models* assume that “the probability of a positive citation from an opinion to a precedent decreases as the doctrine between the two opinions diverges.” (Clark and Lauderdale 2010, 875).

Both studies, upon which we build, ultimately base their model on the proximity model for it better reflects the assumed practice of citing court decisions: “Given the choice between two precedents with identical doctrine and differing directional outcomes, we believe that a justice in this situation would cite both precedents positively.” (Clark and Lauderdale 2010, 875) Moreover, the proximity model better corresponds to the recent shift to model court decisions in a case-space. Thus, we do not diverge from the original studies.

The Czech legal system is not a precedent system *per se* comparable to the common law tradition. However, judges are still required to provide reasons for their judgments and to respect previous decisions. Because the provisions of the Czech constitutional acts are rather vague, the CCC grounds its cases predominantly in its own caselaw. Moreover, the decision which legal sources to cite, which not to cite and how to cite them does not happen haphazardly. In line with the empirical studies on the role of precedent, we assume that judges behave strategically in the sense that they care about the compliance with their decision. Second, the CCC judges find their decisions more persuasive if “legal consistency can be demonstrated across time and place”. Third, the compliance with the CCC judgments is affected both by the lower courts’ behavior as well as the other branches of power. Lastly, the embedding of decisions into the net of caselaw imputes time and effort costs on the CCC judges, who, therefore, have to make the decisions as to which cases to cite and which not to cite (Lupu and Voeten 2012). Citing other CCC decisions enables the CCC justices to explain their conclusions and reasoning as well as to embed the decision within the network of other decisions. It also follows that not all references are the same: the judges can reference other decisions positively, whenever the cited decision and its reasoning is similar, and negative, whenever it is dissimilar.

3.2 Model selection and specification

We now discuss the Clark and Lauderdale (2010) and Clark, Engst, and Staton (2018) models in depth and discuss how we adopted them for the CCC context. Both models share certain theoretical assumptions but diverge on practical implementation.

Both models are based - at least implicitly - on the item response theory (IRT). In psychometrics, the IRT is employed to model “the probability of an examinee responding to an item response given their level of a latent variable(s) of interest” (Wells 2021). The model estimates the probability of answering a question correctly given certain underlying latent variable, the ability of the examinee

for example.³ The IRT enables the researchers to take into account features such as the item difficulty, item discrimination or a pseudo-guessing parameter. Its two main advantages are that given that assumptions of the model have been met, the IRT can efficiently estimate even a large number of parameters using Bayesian Markov Chain Monte Carlo (MCMC), and that the item parameter values (its difficulty or discrimination) do not depend on the person parameter values (their latent ability) (Wells 2021).

Much of the political science on (point) estimating political positions of legislators based on roll-call data (Jackman 2001; Clinton, Jackman, and Rivers 2004) or on their twitter usage (Barberá 2015) is based on the IRT. The extension of IRT to the political space is based on the idea that certain response (a vote in the Parliament or the decision to follow someone on Twitter) is known and that it can be used to estimate the latent variable of the person, such as their political stance. In the latter study, Barberá (2015) estimated the position of political actors as well as ordinary Twitter users based on their decision to follow or not to follow other accounts. In a simple uni-dimensional (from the perspective of the latent variable) without the guessing parameter, the IRT model is as follows:

$$P(u_{i,j} = 1 | \theta_j, \alpha_i, \beta_i) = \frac{\exp(\alpha_i(\theta_j - \beta_i))}{1 + \exp(\alpha_i(\theta_j - \beta_i))}$$

Where $u_{i,j}$ is the item response for the person j on item i , θ_j is the latent variable of interest - the point estimate, α_i and β_i represent the item parameters - its discrimination and its difficulty. The model is a logistic model as it follows the $\frac{\exp(X)}{1+\exp(X)}$ structure of a logistic function. Logically, the closer the ability of the examinee to the difficulty of the item, the more likely is then the correct response. What if, however, the β was defined as θ of another person? That is exactly what the Barberá (2015), Clark and Lauderdale (2010) and Arnold, Engst, and Gschwend (2023) studies do. They rely on the idea that based on knowing the item response, i.e. whether two users followed each other or whether one court decision cited another, we can replace the term $(\theta_j - \beta_i)$ with the distance between the location of two persons, i.e. court decisions, such as the distance between the location of decision i θ_i and location of another decision j ϕ_j equals to their Euclidean distance $\|(\theta_i - \phi_j)\|^2$. The MCMC algorithm then iterates draws from the posterior distribution long enough to estimate the location of all decisions fed to the model.

The two models differ as to what they consider to be the item response. In the Clark and Lauderdale (2010) study, the item response is whether a court decision cited another positively, whereas in the Arnold, Engst, and Gschwend (2023) study, the item response is whether a court decision at all referred to another legal source. Clark and Lauderdale (2010) examine SCOTUS, which, like the Czech Constitutional Court, primarily cites its own decisions (Clark and Lauderdale 2010). They consider only these citations, which they further divide into positive and negative based on their relationship to the cited decision. More specifically, the authors collected all freedom of religion and search and seizure cases and then manually annotated citations among those decisions as either positive and negative. The estimation was based on such an encapsulated, self-referential dataset and on the citations being either positive or negative.

In contrast to that, Arnold, Engst, and Gschwend (2023) study German general courts of lower instances, their focus was not the highest *Bundesverfassungsgericht* but the consistency of caselaw

³For an interesting application to estimate the fairness of examination at law faculties, see Hufeld (2024)

on the state level of the judiciary. Ordinary lower courts typically refer to various legal provisions and decisions of other courts (Arnold, Engst, and Gschwend 2023). Hence, they only use information about references to other legal texts, which can include not only decisions of the same court (as in the case of SCOTUS) but also provisions of any legal regulation or decisions of any court.

The feature selection of the two models is justified by the nature of typical citations on the highest level of the judiciary - SCOTUS or constitutional courts - and on the lower levels of ordinary judiciary. The caselaw citations of SCOTUS or constitutional courts are typically self-referential. Citations of laws typically refer to vague constitutional provisions, which will be typically shared by cases on both extremes on a latent dimensions, and as such do not contain any information as to the location of the decision on the latent dimensions. In contrast, a lower court decision will typically cite decisions of higher courts and very precisely formulated legal provisions. In practice, a conservative freedom of speech decision will refer to the same vague and broadly-formulated provisions as a progressive freedom of speech decision, whereas strict lower court defamation decisions will altogether refer to different legal provisions than lenient defamation decisions.

Clark and Lauderdale (2010) also make use of data on the estimated ideology of the SCOTUS justices to globally identify the model. We do not have such data as the CCC justices generally do not display such clear political positions. Additionally, the CCC is made up predominantly of appointees of the same president at any given time (Smekal and Vyhnanek 2020, 128). In contrast to that, Arnold, Engst, and Gschwend (2023); Barberá (2015) do not globally identify their models at all.

Our model merges both approaches. Instead of Poisson distributed citation counts as in Arnold, Engst, and Gschwend (2023), we employ Bernoulli distributed variable to capture whether certain legal source was cited or was not cited by a CCC decision. Firstly, we are able to narrow our data to include only positive citations in Clark and Lauderdale (2010) fashion. Secondly, as long as the citing behavior of different judge rapporteurs varies, the estimates may be biased. Arnold, Engst, and Gschwend (2023) posit in their appendix that “Recording only whether a court refers to a legal source or not is quite a strong assumption. It seems much more realistic to also think about how often a court is citing a source. While a decision considers a dissenting legal source, it might refer to it only once or twice. But a legal source that is relevant will be referred to much more often.” However, some CCC decisions cite a key source once and then proceed to interpret it in extensive fashion, others mention a certain legal source multiple times even within one paragraph. Therefore, we do not think that “recording only whether a court refers to a legal source” is that strong of an assumption within the CCC context – especially given the fact that we are able to filter only those references that are truly positive and relevant for the decision.

Therefore, our final model is as follows. We have a $i \times j$ decision-source matrix $\mathbf{Y}$, where Y_{ij} equals 1 if the opinion i positively cites a legal source j and 0 if it does not cite it at all. We use a Bernoulli distribution as the probability model for our citation data. Similarly to Arnold, Engst, and Gschwend (2023) and Clark and Lauderdale (2010), the model rests on three assumptions: (1) Each decision i has a fixed location along a uni-dimensional space, (2) the decisions are more likely to positively refer to the same legal source the closer they are in the case-space⁴, and (3) each

⁴Here we are not sure about the phrasing of the second assumption in Arnold, Engst, and Gschwend (2023): “A written decision will refer to a legal source more often the closer their locations in this common case-space.” As long as we are not placing the legal sources within the same case-space, which is not possible given that legal acts are not part of a case-space on the same level as cases but rather as the cut points through that case-space, then such an

decision has a positive probability to refer to any legal source.⁵ Where ϕ is the standard normal cumulative distribution function (i.e. a probit link), the probability of a decision i citing a legal source j is

$$Pr(Y_{ij} = 1 | \alpha_j, \beta_i, \gamma, \theta_i, \phi_j) = \text{Bernoulli}(\Theta_{ij})$$

$$\Theta_{ij} = \Phi(\alpha_j + \beta_i - \gamma ||\theta_i - \phi_j||^2)$$

Building on the aforesaid IRT model of Barberá (2015) and of Arnold, Engst, and Gschwend (2023), the parameters θ_i is the location of decision i , ϕ_j is the location of legal source j , α_j expresses the authority of legal source j , i.e. its propensity to be cited more or less likely than others in general due to its overall relevance, β_i captures decision-specific differences. The model is then estimated using the Bayesian programming language Stan implemented in R as RStan (Brooks et al. 2011).

3.3 Data

To apply the model, we used data from the CCC database (Paulík 2024) and metadata provided by the Beck-online database. The official CCC database *NALUS*⁶ consists of the texts of all CCC decisions and accompanying metadata (such as dates, judge rapporteur, registry items, relevant legal provisions).

3.3.1 Data Sources

To identify relations between the decisions, we use metadata that was kindly provided by the Czech legal information system *Beck-online*⁷ owned by C. H. Beck publishing house. In this respect, we follow the approach of both Clark and Lauderdale (2010) (using Shepard’s citation data) and Arnold, Engst, and Gschwend (2023) (using data by the German legal database Juris). Beck-online captures court decisions cited by another decision, together with the “quality” of the citation, i.e. capturing the court’s treatment of the decision cited. This type of metadata is similar to *Shepard’s Citations* used by the U.S. legal databases but pretty rare in the European markets.

Beck-online uses a total of eight types of quality⁸ similar to the Shepard’s typology (Spriggs and Hansford 2000, 330); these are displayed to users in the metadata of the citing and cited decision. Citation metadata capture only decisions cited by the court itself and only as a part of court’s argumentation. That approach excludes decisions mentioned solely by parties or in procedural context only. One type of quality indicates solely that the decision is cited by a dissenting (or concurring) opinion, which leaves us with seven substantive qualities describing the attitude of the majority opinion. To simplify the input data in the vein of Clark and Lauderdale, as well as other scholars (cf. Spriggs and Hansford 2000, 335), we flatten these seven qualities into two: positive or negative.

assumption does not make sense. Our assumption is more appropriate in the sense that it captures the essence of the model: if two decisions cite the same source, then they are likely to be closer to each other in the case-space.

⁵We do not include the Clark and Lauderdale (2010) assumption that missing citations are missing at random. It should be the case that the manually annotated data from Beck should also include citations that the CCC may have forgotten to include. Even then the assumption would very likely hold.

⁶Available at: <https://nalus.usoud.cz/>

⁷Available at: <https://www.beck-online.cz/>

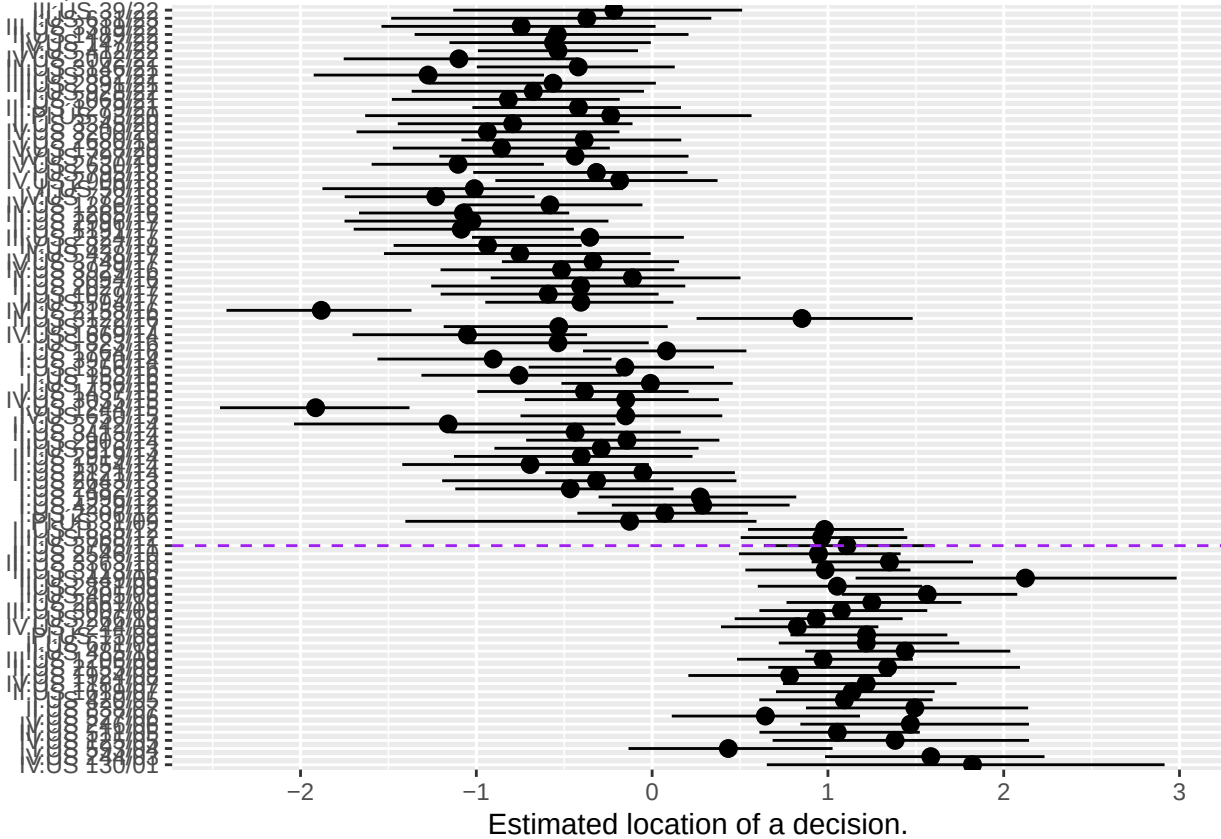
⁸In addition, some citations may not be annotated with quality yet.

Consistently with the Shepard’s approach (cf. [Spriggs and Hansford 2000, 331](#)) we decided to code *distinguishing* quality as negative treatment. The reasons were twofold: first, it helped with the small share of negative citations in the dataset. Second, distinguishing can be a way of getting past uncomfortable caselaw without disagreeing with it explicitly. In that respect, distinguishing is often a concealed rejection.

3.3.2 Narrowing the case-space

To narrow down cases from within one case-space, we have relied on two types of information. First, we relied on the metadata of the CCC database, which includes information on the subject matter as well as area of law for each decision. As we show later in the discussion on the consumer cases, this has proven to be an insufficiently rough way to filter the decision. Even when we defined our subject matter and area of law narrowly, the resulting data would include differing sub-topics from different “case-spaces”. As such, the model would capture the differences in these sub-topics rather than differences between the positions of the decisions themselves. Second, to narrow the cases further, we selected a handful of representative cases for the topic we were interested in. The CCC database includes the popular name for all judgement and the Beck-online data include the legal sentences for majority of the decisions on merits. We transformed the popular names as well as the legal sentences into doc2vec embeddings. We then searched the whole CCC database for decisions with the highest cosine similarity of the popular names in the vein of Arnold, Engst, and Gschwend ([2023](#)).

Moreover, when making the decision which data to include, we faced the issue of legal reforms (recodifications) having taken place in Czechia. If the data on cited legal acts are included in the data fed to the model, the model captures legal reforms rather than underlying shifts in judicial practice. To show this issue in practice, we run the model on custody cases. Until 2013, custody cases were governed by the old 1964 Civil Code. The shift in ?? corresponds to adoption of the 2012 Civil Code. The red vertical line indicates the first decision that has been made after the 2012 Civil Code came into force. Practically all decisions thereafter fall to the one side of the x-axis, whereas the old decisions fall to the other side.



Therefore, if the subject at question spans longer time in which legal changes took place, it is reasonable to exclude the data on legal acts in the model because otherwise the model captures the legislative rather than judicial development. Because the data on GDR, for example, spans a long time, the underlying laws have changed and with it the references the CCC makes. Therefore, for subjects spanning a longer time, we exclude references to laws. In contrast to that, the caselaw on consumer concerns only the 3rd term of the CCC. Therein, it makes sense to include references to law as well in the vein of Arnold, Engst, and Gschwend (2023).

After narrowing the data sufficiently, our model should be fit with data within one case-space and should be robust to legal reforms.

3.3.3 Descriptive statistics of the data

The GDR data span multiple decades; therefore, only references to CCC caselaw are left in the model. As such, the subset of data on the generally binding regulation contains 103 unique citing decisions and 263 cited decisions.

Although the general subject matter tag “consumer” contains 26 unique citing decisions that all originated during the 3rd term of the CCC, we found that the first rounds of the model were capturing different sub-topics within the consumer subject matter. Therefore, the data had to be split into two further sub-topics. Broadly speaking, the first subtopic concerns non-pecuniary damage and the second subtopic concerns mostly various issues with consumer contract law.

The data on effective investigation contains 14 and the model returns 3 decisions whose 95 %

Table 1: Descriptive statistics for the data used for fitting the models.

Subject Matter	Observations	Mean Citations	Median Citations	SD Citations
GDR	103	6.7	4.0	8.7
Consumer Protection	26	19.8	17.0	12.2
Effective Investigation	14	13.9	11.5	8.7

credible interval falls outside 0.

Tab. 1 presents summary statistics for the data used for fitting the models, more specifically, the number of observations, average as well as median number of citations per each decision, and the standard deviation of thereof are included.

4 Results

Even if we control for the legal reforms, we seem to capture the underlying legal topic. In other words, just using simple metadata of the CCC decisions, we are not able to narrow down the case-space sufficiently. To test that and to potentially narrow down the case space, we employ a similar method to Arnold, Engst, and Gschwend (2023). Arnold, Engst, and Gschwend (2023) convert the titles of a couple of pre-selected decisions into tf-idf and find the most similar decisions using cosine similarity to delimit the case-space as narrowly as possible. The results of running to model on consumer cases reveal that there are four decisions to the left of 0. We convert the popular names of these four decision as well as the other 7000 decisions that have a popular name into dense doc2vec vectors (Le and Mikolov 2014), which is a more modern method to represent text as a vector than the tf-idf matrix based on the bag-of-words approaches.

The results from finding the most similar decisions based on the cosine similarity of the doc2vec vector representations of the popular titles of decisions reveal that the 3 closest decisions to either of the four highlighted decisions are the remaining three decisions. Therefore, simply inputting all cases tagged with the consumer tag in the CCC database does not suffice to delimit the case-space, on which the model would be able to capture caselaw consistency. Mohammadi et al. (2024) have indeed found that citation networks can be used to improve topic modelling of caselaw of the European Court of Human Rights.

However, when we apply the model to the subsetted data of the four decisions that – according to the filtering based on the cosine similarity – fall within the same case-space, we find that the cases seem to be consistent. None of the decisions is statistically significantly different from 0. Therefore, the conclusion would be that there is no inconsistency among the decisions. A close reading of the four decisions indeed corroborates the model’s conclusion. All four decisions have been granted on similar grounds and concern the usage of evidence in proceedings concerning compensation for non-pecuniary damage for a delayed flight. We argue that in this narrowly constructed case-space, the model correctly captures the consistency of CCC caselaw.

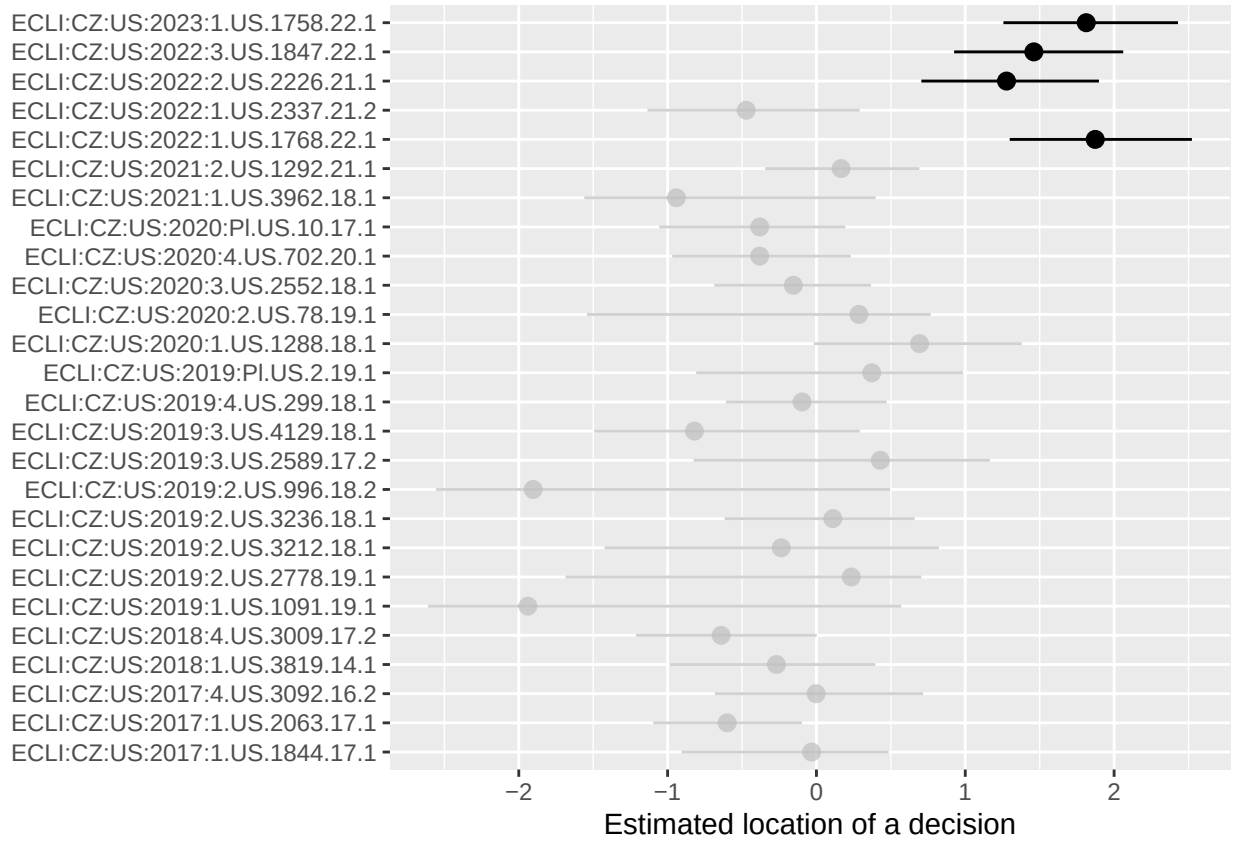


Figure 1: The results of the fitted model on the consumer cases. The four decisions with significant $\theta < 0$ are highlighted.

Table 2: A table showing the 10 most cosine similar cases to the four statistically significant consumer law cases whose position was estimated to be negative. The highlighted rows are those, in which the most similar decision was one of the other three from the four.

Case	Similarity	Rank
ECLI:CZ:US:2022:2.US.2226.21.1		
ECLI:CZ:US:2022:3.US.1847.22.1	0.971	1
ECLI:CZ:US:2023:1.US.1758.22.1	0.921	2
ECLI:CZ:US:2022:1.US.1768.22.1	0.908	3
ECLI:CZ:US:2022:4.US.2620.21.1	0.820	4
ECLI:CZ:US:2009:2.US.1191.08.1	0.783	5
ECLI:CZ:US:2014:2.US.3863.13.1	0.753	6
ECLI:CZ:US:2000:4.US.548.99	0.753	7
ECLI:CZ:US:2013:3.US.2365.11.1	0.750	8
ECLI:CZ:US:2013:4.US.3115.12.1	0.734	9
ECLI:CZ:US:2016:4.US.3973.13.1	0.733	10
ECLI:CZ:US:2022:1.US.1768.22.1		
ECLI:CZ:US:2022:3.US.1847.22.1	0.939	1
ECLI:CZ:US:2022:2.US.2226.21.1	0.908	2
ECLI:CZ:US:2000:4.US.548.99	0.845	3
ECLI:CZ:US:2023:1.US.1758.22.1	0.843	4
ECLI:CZ:US:2009:2.US.1191.08.1	0.775	5
ECLI:CZ:US:1997:3.US.218.97	0.747	6
ECLI:CZ:US:2007:1.US.626.06	0.744	7
ECLI:CZ:US:2022:3.US.2729.21.1	0.740	8
ECLI:CZ:US:2014:1.US.1085.13.1	0.739	9
ECLI:CZ:US:2013:1.US.3270.12.1	0.736	10
ECLI:CZ:US:2022:3.US.1847.22.1		
ECLI:CZ:US:2022:2.US.2226.21.1	0.971	1
ECLI:CZ:US:2022:1.US.1768.22.1	0.939	2
ECLI:CZ:US:2023:1.US.1758.22.1	0.910	3
ECLI:CZ:US:2022:4.US.2620.21.1	0.801	4
ECLI:CZ:US:2009:2.US.1191.08.1	0.787	5
ECLI:CZ:US:2000:4.US.548.99	0.787	6
ECLI:CZ:US:2019:4.US.2287.18.1	0.752	7
ECLI:CZ:US:2013:3.US.2365.11.1	0.747	8
ECLI:CZ:US:2014:2.US.3863.13.1	0.739	9
ECLI:CZ:US:2022:3.US.2729.21.1	0.738	10
ECLI:CZ:US:2023:1.US.1758.22.1		
ECLI:CZ:US:2022:2.US.2226.21.1	0.921	1
ECLI:CZ:US:2022:3.US.1847.22.1	0.910	2
ECLI:CZ:US:2022:1.US.1768.22.1	0.843	3
ECLI:CZ:US:2009:2.US.1191.08.1	0.739	4
ECLI:CZ:US:2014:2.US.3863.13.1	0.722	5
ECLI:CZ:US:2022:4.US.2620.21.1	0.722	6
ECLI:CZ:US:2012:1.US.216.09.1	0.719	7
ECLI:CZ:US:2012:3.US.2142.09.1	0.715	8
ECLI:CZ:US:2016:1.US.2394.15.1	0.715	9
ECLI:CZ:US:2013:4.US.3115.12.1	0.712	10

4.1 Generally binding regulations

Out of the 103 decisions that were fed into the model, 60 were estimated as significantly different from 0 as their 95 % credible intervals did not cross 0. Fig. 2 presents the estimates of the location of the decision.

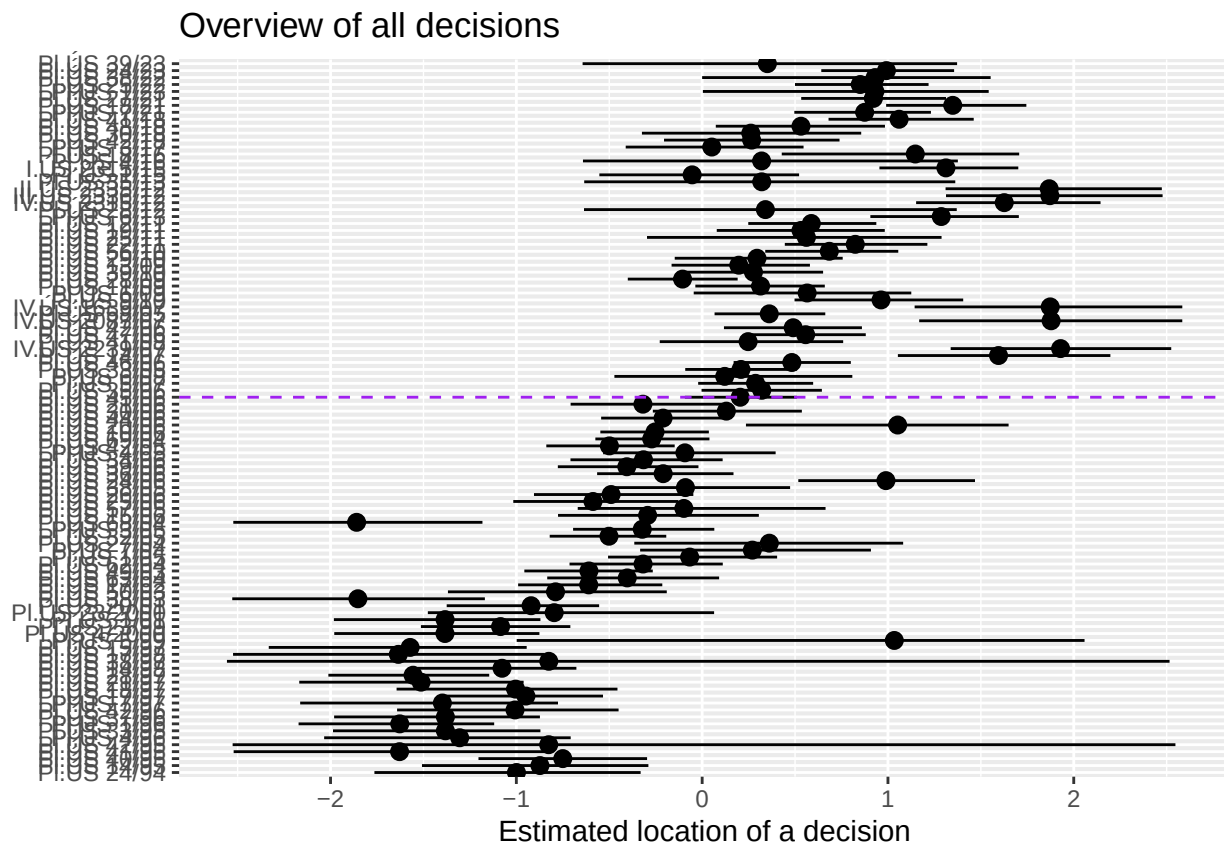


Figure 2: Overview of estimated positions of all the GBR decisions. The Jirkov case Pl. ÚS 45/06 is highlighted with a red line.

We used the cosine similarity method to verify whether the model does not capture an underlying difference in topics. Table ?? shows that that is not the case. While most cases that appear in the table are grey, i.e. from the same case-space, the top positions for each statistically significantly different decision from 0 are not predominantly occupied by the pink decisions, i.e. those from the same side from 0.

We can observe that the average position of cases shifts around 2005. In the Czech legal scholarship, the *Jirkov* judgement is often considered to be the landmark case that heavily shifted the CCC caselaw in the field of generally binding resolutions, widening the regulatory powers of municipalities (see above). However, a closer look at the fig. 2 reveals that the shift in the caselaw may have happened earlier. The landmark *Jirkov* judgement is highlighted. The shift precedes the landmark decision by a little bit, supporting the part of scholarly literature that describes the *Jirkov* judgement as only the final confirmation of the shift (Kosař and Vyhnánek 2023). The only decision estimated at the minus side of the x-axis (REFORMULOVAT) after the shift⁹ was, regarding the

⁹Judgment of the CCC of 6 June 2006, Pl. ÚS 68/04.

facts of the case, a kind of a remnant of the 1990s caselaw where the CCC struck down multiple GBRs banning communist and Nazi propaganda.

We thus posit that in the case of GDRs, the caselaw shift did not occur at once. Rather, a landmark decision explicitly confirmed a gradual process that had already been ongoing for a certain amount of time.

Moreover, looking at the consumer and XXX data, we find little to none inconsistency of the CCC caselaw across its chambers. That may have two potential explanations. Firstly, the model does not capture the inconsistency even with the data we provide it. While that may be the case, we have done our best job to narrow down the case-space as much as possible and to feed the model only with meaningful positive citations. The second explanation is more reasonable. As we discussed earlier, we were faced with the decision whether to include decisions on merits or also those on admissibility.

The second alternative explanation that could not have been captured with our model is that the judges strategically choose which cases to reject on admissibility and which to grant on merits. Because – unlike cases rejected on merits – the cases rejected on admissibility are not binding for the future, a CCC justice who does not harness majority for their desired disposition is motivated to decide the case on admissibility rather on merits so as not to make the opinion they disagree with binding. Similarly, rather than creating inconsistency via two decisions on merits, the justice may either dismiss the complaint as inadmissible or activate the plenary procedure for caselaw unification. Therefore, the inconsistency shows up either in the form of a case with similar facts being granted by one judge and being dismissed on admissibility by another, or in the form of the actual plenary procedure for unification of the CCC caselaw. Neither mechanism can be captured by our model. Even if we included the inadmissibility decisions, they by their nature usually contain completely different and much sparser references to caselaw, as well as to legal provisions. While our conclusion is not a groundbreaking one, it is one backed by evidence and confirms what Czech legal scholars have been saying.

4.2 Consumer

No matter how hard we try to filter the decisions into one case-space, the model always picks up a very small sub-topic of the dataset. The first run uncovered that a subset of the consumer protection cases deals with the extraordinary appeal to the Supreme Court. The second run on all consumer cases uncovered that there are three cases that concern non-pecuniary damage for delays in consumer transportation. The left side of fig. 3 shows that all the highlighted decisions are those that concern delays, the right plot reveals that when the model is run only on these four cases, neither differs from zero.

There are a couple of potential explanations. Firstly, it may be that we have still failed to narrow the case-space down sufficiently. Were this true, it would then mean that some of the case-spaces are left with only two to four decisions, too few to base any meaningful model on. Secondly, because when we heavily filtered the data to include only the decisions truly falling within the small case-space, none of the decisions in the smaller case-spaces appeared inconsistent with the rest, it may just as well be that the decisions on merits of the CCC are consistent.

That, we believe, has broader implications for the issue of measuring inconsistency. We posit that

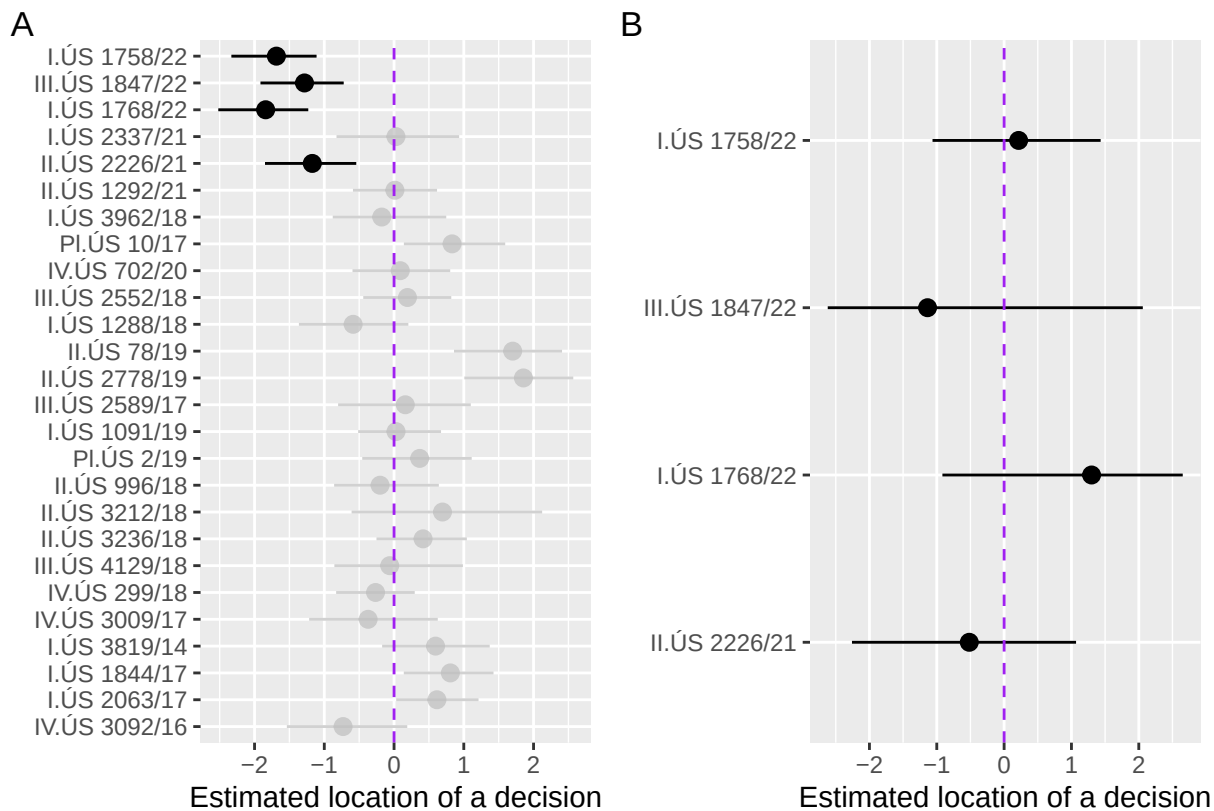


Figure 3: A plot showing the fitted model on consumer cases. The plot A is fitted on all 26 cases and it highlights those that were statistically significantly lower than 0. The plot B then fits the model with only the four cases concerning non-pecuniary damages for consumer transport. None of the cases significantly differs from 0.

were a clear inconsistency to occur between decision merits, the CCC has a mechanism in place to address it before it occurs. Secondly, the formal definition of caselaw relies on the counterfactual comparison between decisions on the same facts. While in real life, we cannot observe the counterfactual, we posit that even finding similarity of facts of a case, in order to make the cases comparable, may be a higher threshold than we had assumed. In other words, even when legal scholarship concludes that some decisions are inconsistent, unless explicitly stated in the decisions themselves, it could just be that the facts of the case are subtly yet sufficiently different.

Lastly and most importantly: given the last two points, we posit that the inconsistency we are expecting to observe in the CCC caselaw actually occurs between decisions on merits and decisions on admissibility. Earlier, we argued that simply comparing the “winrates” across judges does not suffice to reveal inconsistency. However, if the case-space is narrowed down sufficiently, thus there is no systemic skew stemming from different facts of the case or strategic case assignment (Fjelstul 2023), then a simple comparison of proportions of cases granted on merits and rejected on admissibility or on merits suffices. Figure 4 reveals that indeed the win rates in the consumer cases vary quite a lot. The caveat is that there are too few cases to draw any strong conclusions – yet the disparity is quite stark even at the low number of cases. For example, out of the two judges with the most cases, Jan Filip has granted around 50 % of consumer cases, whereas David Uhlíř has granted only 18 %. Moreover, we can see that simply looking all cases does not paint the whole picture within each case-space.

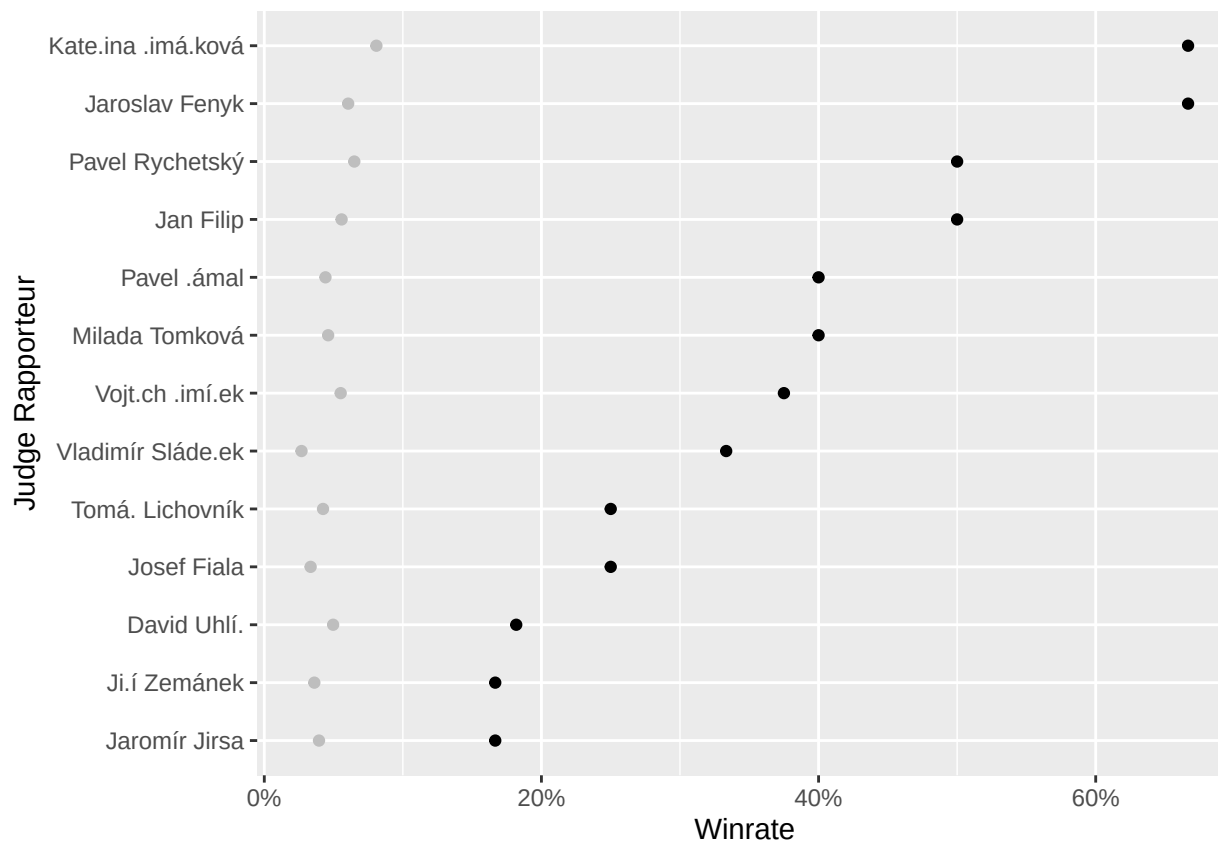


Figure 4: A table comparing the winrates among the judge rapporteurs. The black dot are only consumer cases, gray dot are all cases.

4.3 The Right to Effective Investigation

The right to effective investigation in criminal proceedings is an area of law that has undergone a caselaw development in the last two decades, mainly following the caselaw of the European Court of Human Rights. Blažková (2016) uncovers and describes the development. She argues that the CCC at first approached the issue with restraint; it held that because of the separation of powers, it's not the CCC's task to meddle with criminal proceedings and that there is no human right that another person be criminally prosecuted. However, a change in approach towards the right to effective investigation could be observed in two out of four chambers of the third term of the CCC. These judgements cited evolving caselaw of the ECHR, including judgements made against the Czech Republic. Blažková (2016), p. 10-11 criticizes that this shift was not made by the appropriate plenary procedure, but rather by senate judgments. The article handily singles out and classifies relevant decisions as either supporting or undermining the right to effective investigation.

Therefore, an inconsistent treatment of effective investigation constitutional complaints that has been mapped by a legal scholar occurred in the years from 2013 until writing of the article in 2016. We filtered only the decisions analysed by Blažková (2016) and then grouped them into two groups. The variable *article* takes up either value “restraint” or “active”, depending on how the author classified the decision in question. We then compared the results of the model against claims of Blažková.

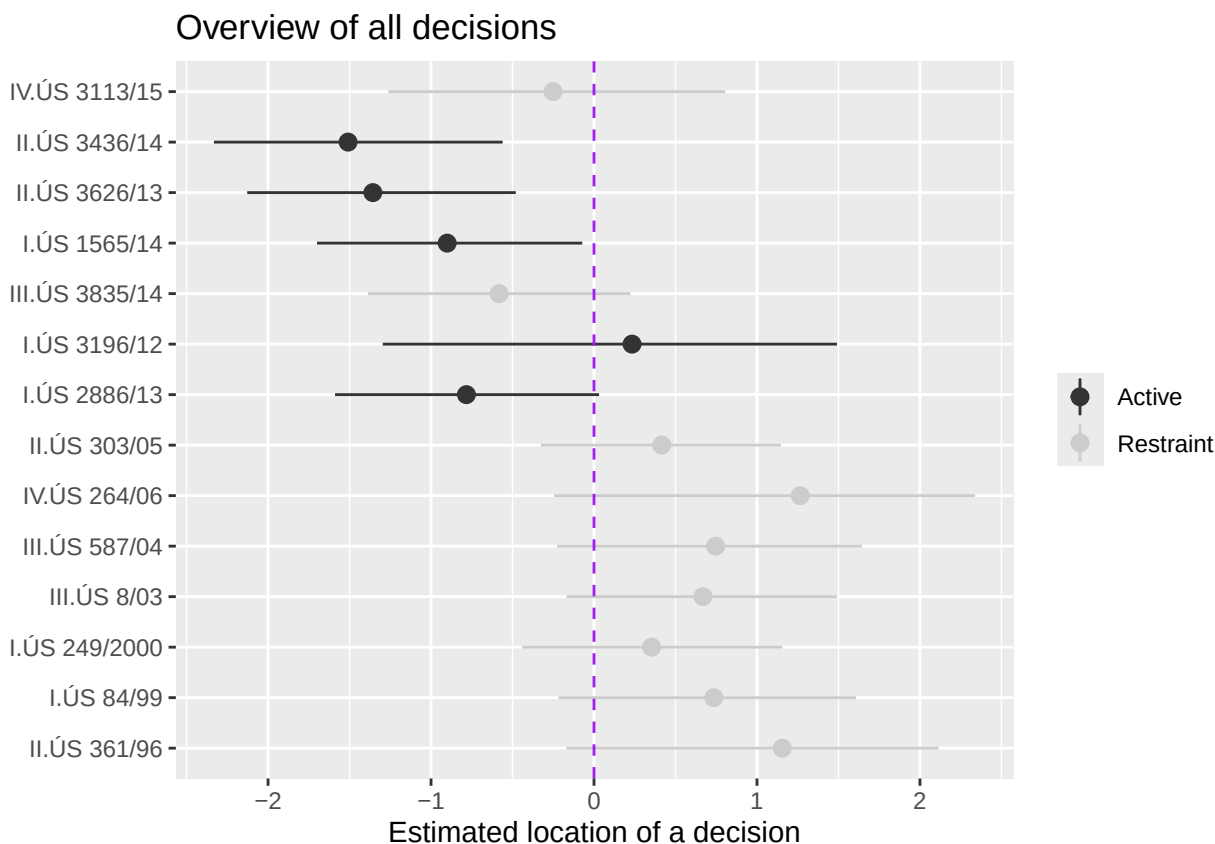


Figure 5: The effective investigation plot.

Fig. 5 reveals that while our model does not capture the underlying classification perfectly, it

reasonably captures the trend proposed by Blažková. The highlighted cases are those labeled as progressive. We can see that three out of the five significantly differ (at 95 % credible interval), one almost differs, and one does not, whereas the restrained decisions fall either to the middle or to the right of 0. While a new Criminal code was passed in 2009, it entered into force only in 2010 and given the typical length of proceedings, all the decisions in our dataset still refer to the old Criminal code. Therefore, the conclusion is also robust to the criminal law reform that occurred in 2010.

Conclusion

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